

Performance Report

Evaluation

Installation	Equipment	Type	No.
APL HOUSTON	MAIN ENGINE	10S90ME-C	DML0105356

Readings Included

	Date	Voyage / title	Remarks
1	2025-11-11	04FLZW1MA	ENROUTE FROM SGSIN TO CGPNR

Main indications (all Readings)

Reading	1
Description	Most recent
Fuel oil saving potential	●
Light running	●
Fuel index ECU	●
Firing press. AVG	●
Compression press. AVG	●
Scav. air press. RECEIVER	●
Exh. gas temp. TC in AVG	●
Exh. gas temp. TC out AVG	●
TC speed AVG	●
Spec. FO consumption	🔴 Low

Indications (for Reading 1)

Description	Indication	Comments / recommendations
Spec. FO consumption	🔴 Low	Wrong measurement. / Wrong engine power used for calculation.
Cool. water temp. CYL out AVG	🔴 Low	Setting of temp. controller low.
MIP deviation CYL 1	🔴 High	MOP: load parameters [offset high or low load]. / Indicator cock or measuring device problem. / TDC detection or evaluation. / Unstable load.

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Optimization potential

Quantification						
Influenced by	Fuel oil equiv.	Thermal load	Proposed actions to achieve the saving / reduction			
Firing pressure by injection	0.0 g/kWh	0 °C				
Firing pressure by compression ratio	0.0 g/kWh	0 °C				
Firing pressure by scavenge air pressure	0.0 g/kWh	0 °C				
ENGINE ROOM ventilation	0.0 g/kWh	0 °C				
Suction press. air TC in	0.0 g/kWh	0 °C				
Press. drop accross SAC	0.0 g/kWh	0 °C				
Scav. air : water temp. SAC in	0.3 g/kWh	1 °C	Reduce SAC water intlet temperature to 29.2 °C to reduce scav. air temp. by 3 °C.			
Scav. air : temp. RECEIVER	0.0 g/kWh	0 °C				
Press. drop accross ENGINE	0.0 g/kWh	0 °C				
Exh. gas press. TC out	0.0 g/kWh	0 °C				
Light running	0.0 g/kWh	-				
Fuel oil visc. ENGINE in	0.0 g/kWh	0 °C				
	Fuel oil equiv.	Thermal load	Cyl. lub. oil			
TOTAL (specific)	0.3 g/kWh	-				
TOTAL (of current)	0.2 %	-		CO2 (emission)	SOx (emission)	
TOTAL optimization potential	0.15 t/d	2 °C		0.5 t/d	1 kg/d	
	Cost					
Fuel oil price						
TOTAL saving potential						

Consumption

Measured / calculated values			
Indication	Fuel oil		Cylinder lub oil
Specific consumption reference	168.9 g/kWh		0.70 g/kWh
Specific consumption ISO / MEAS	165.2 g/kWh		0.77 g/kWh
Consumption per hour day *	3516 kg/h	84.4 t/d	15.7 kg/h
Consumption nautical *	200 kg/nmi		
* Measured values, uncorrected			

Injection timing and compression ratio adjustment

Cylinders												
Settings		1	2	3	4	5	6	7	8	9	10	
Proposed adjustments to reach target firing press., positive value means increase of pressure:												
Compr. ratio offset [-]		-0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
pmax offset [bar]		0.0	0.0	-0.1	0.0	-0.1	0.0	0.0	-0.1	0.0	0.0	0.0
Common compr. ratio offset [-]	1.7											
Common pmax offset [bar]	-4.2											

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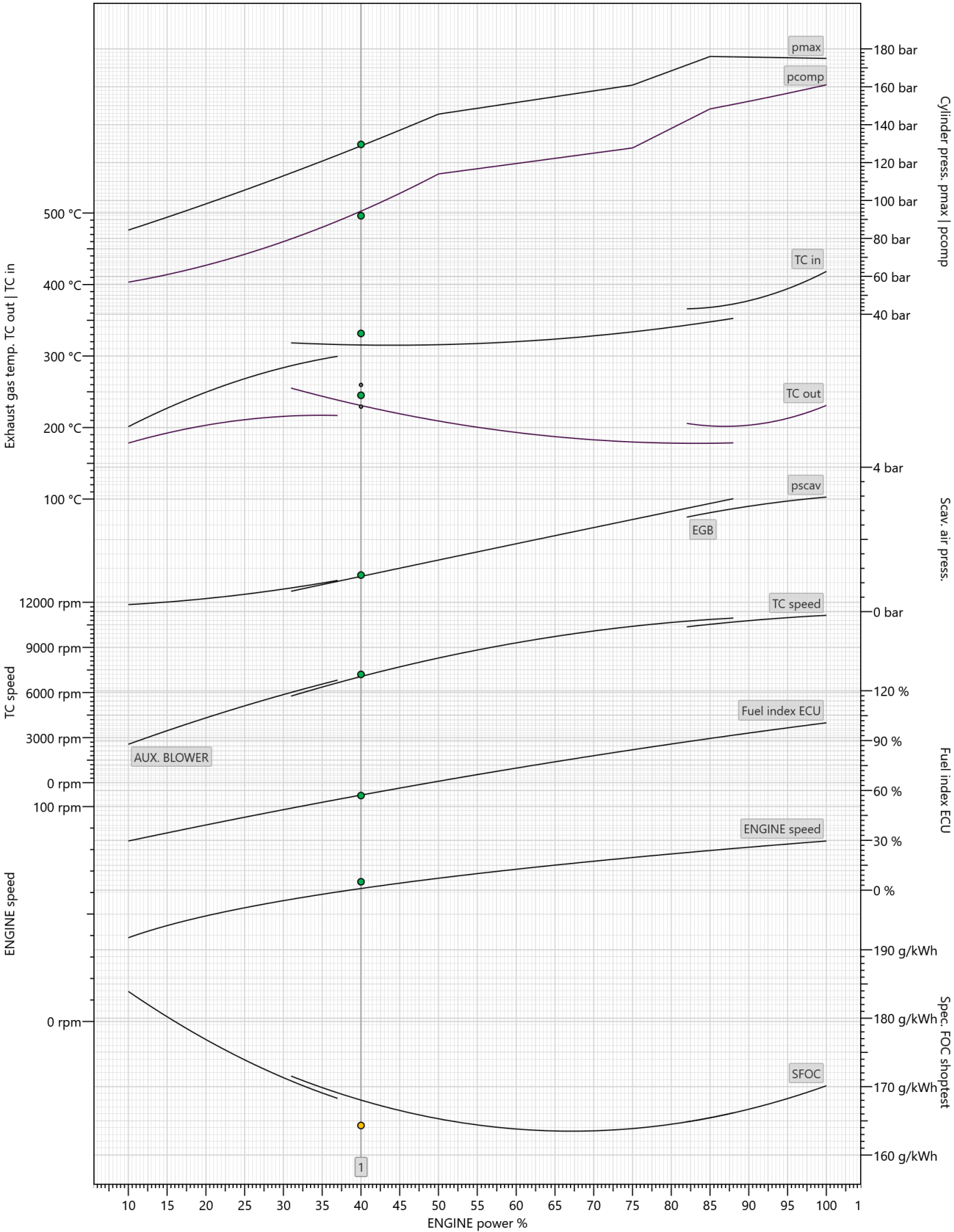
Input plausibility

Value	Comment
Compressor evaluation	
Exhaust gas pressure Manifold	
PTI	
PCO	
TTI vs TTO	
TCO	
Minimum required data	
Minimum required data II	
Missing data	
Missing data II	

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Performance Curves

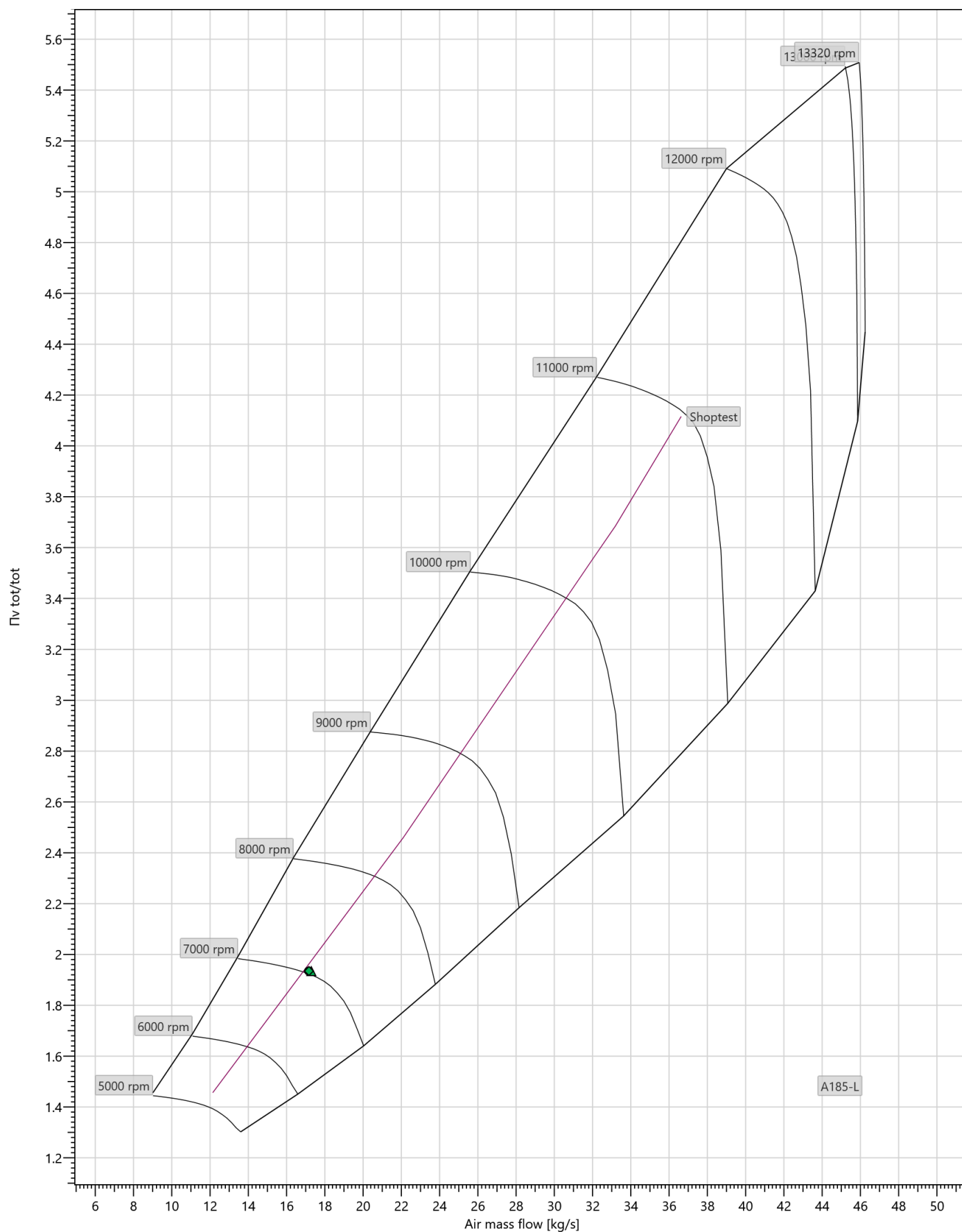
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COMPRESSOR map

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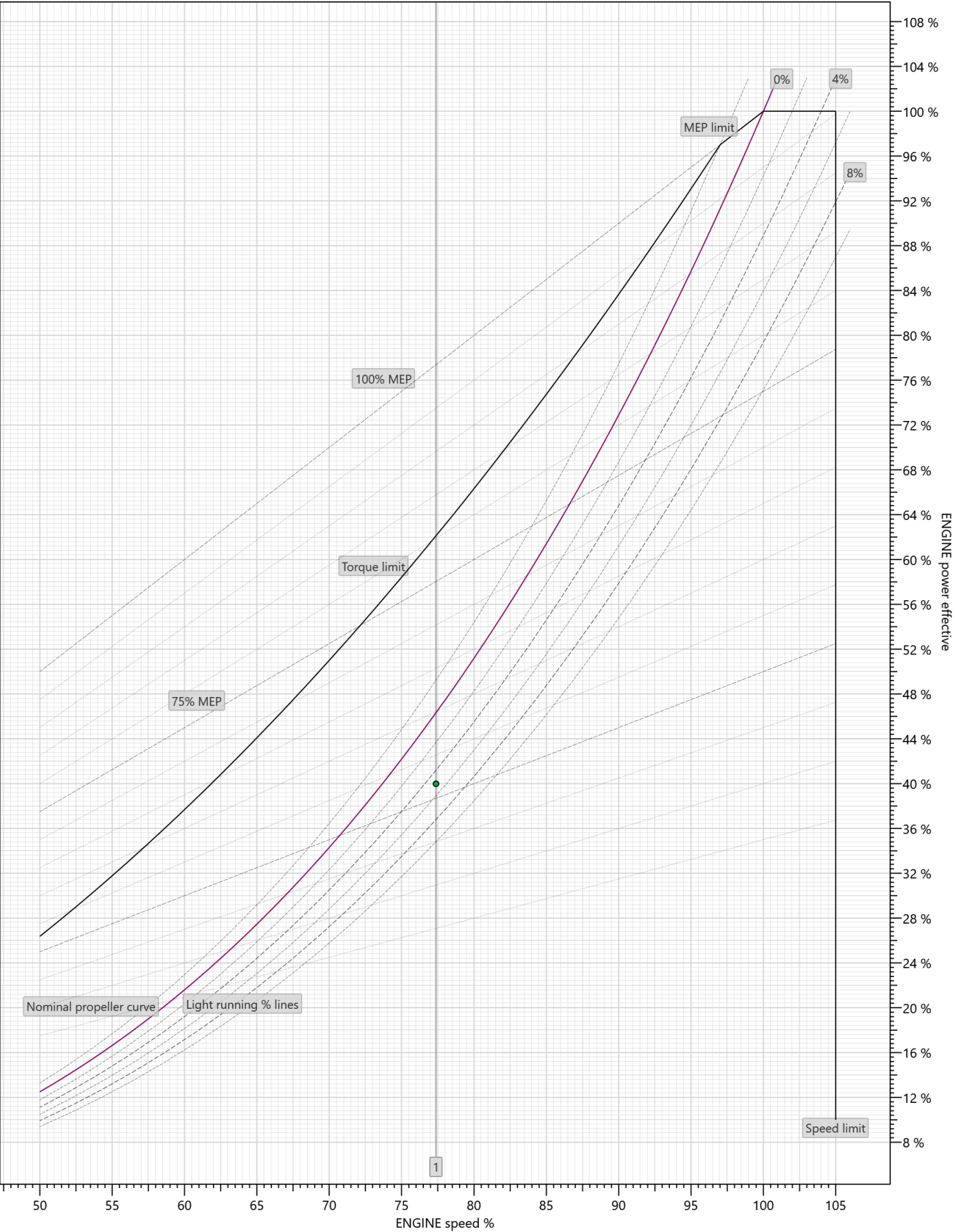


1 2025-11-11 04FLZW1MA		
● TC 1	▲ TC 2	◆ TC 3

Performance Report

Load Diagram

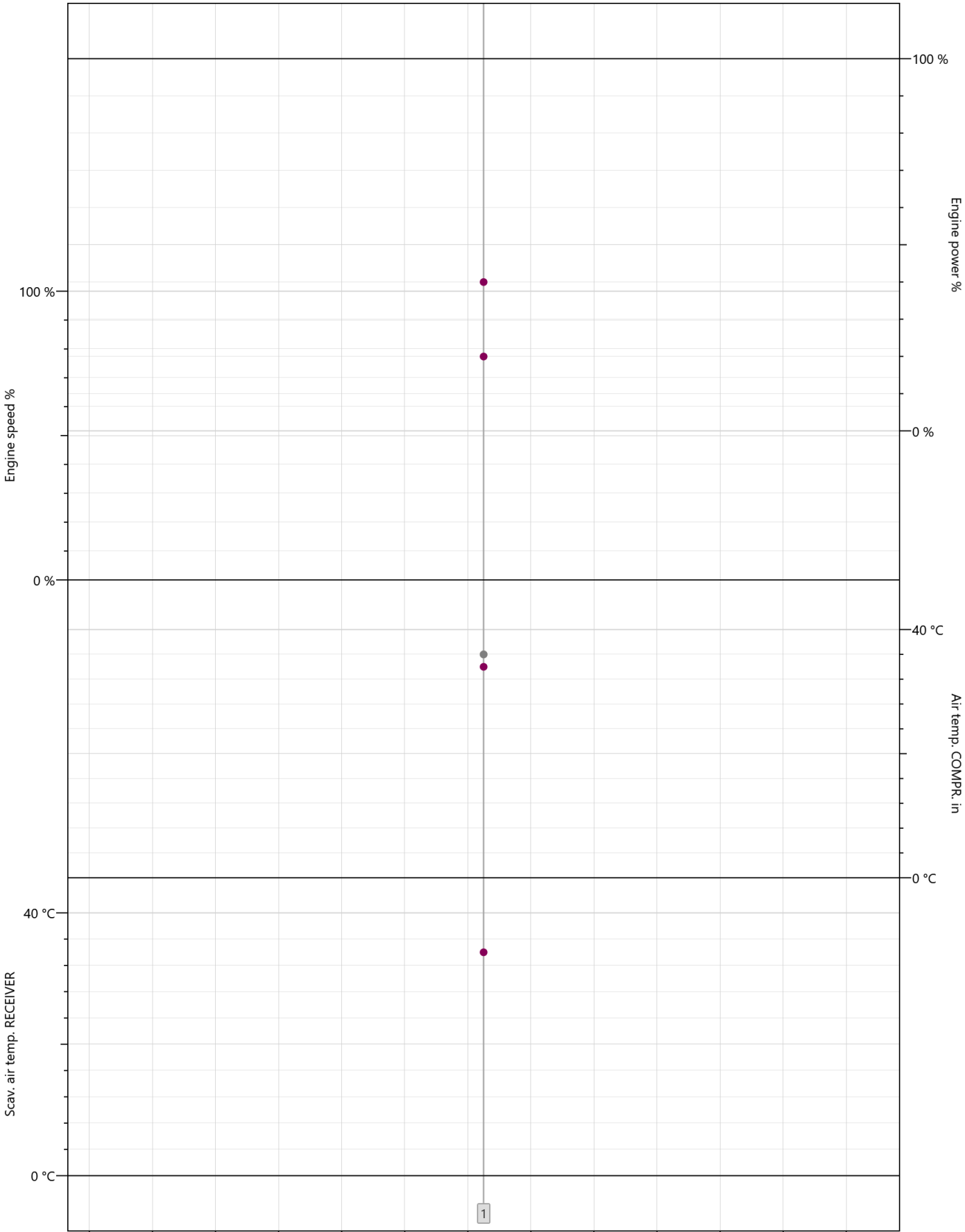
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Trend: LOAD and AMBIENT

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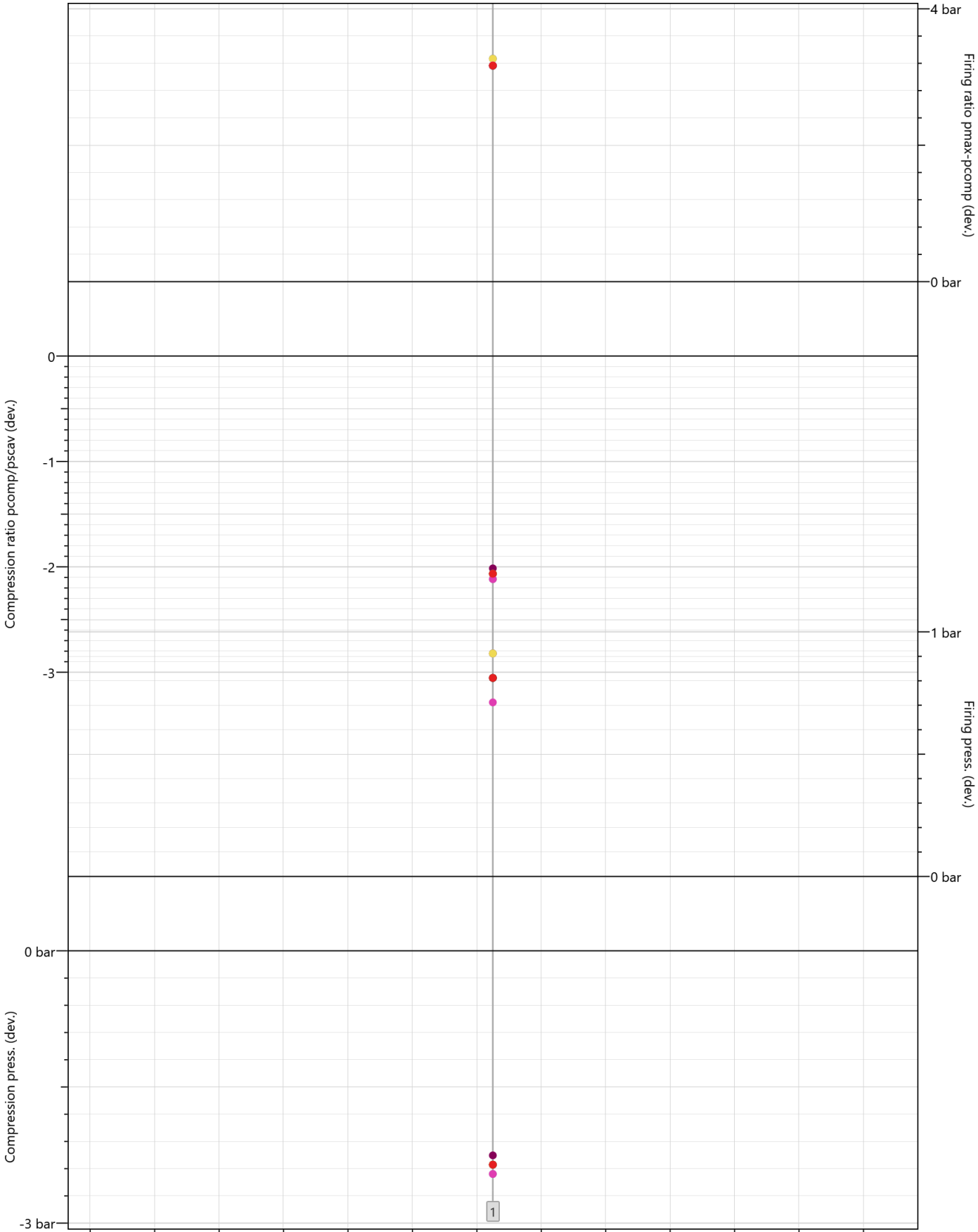


TC1	TC2	TC3	AVG	RECEIVER
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Trend: COMBUST. CHAMBER

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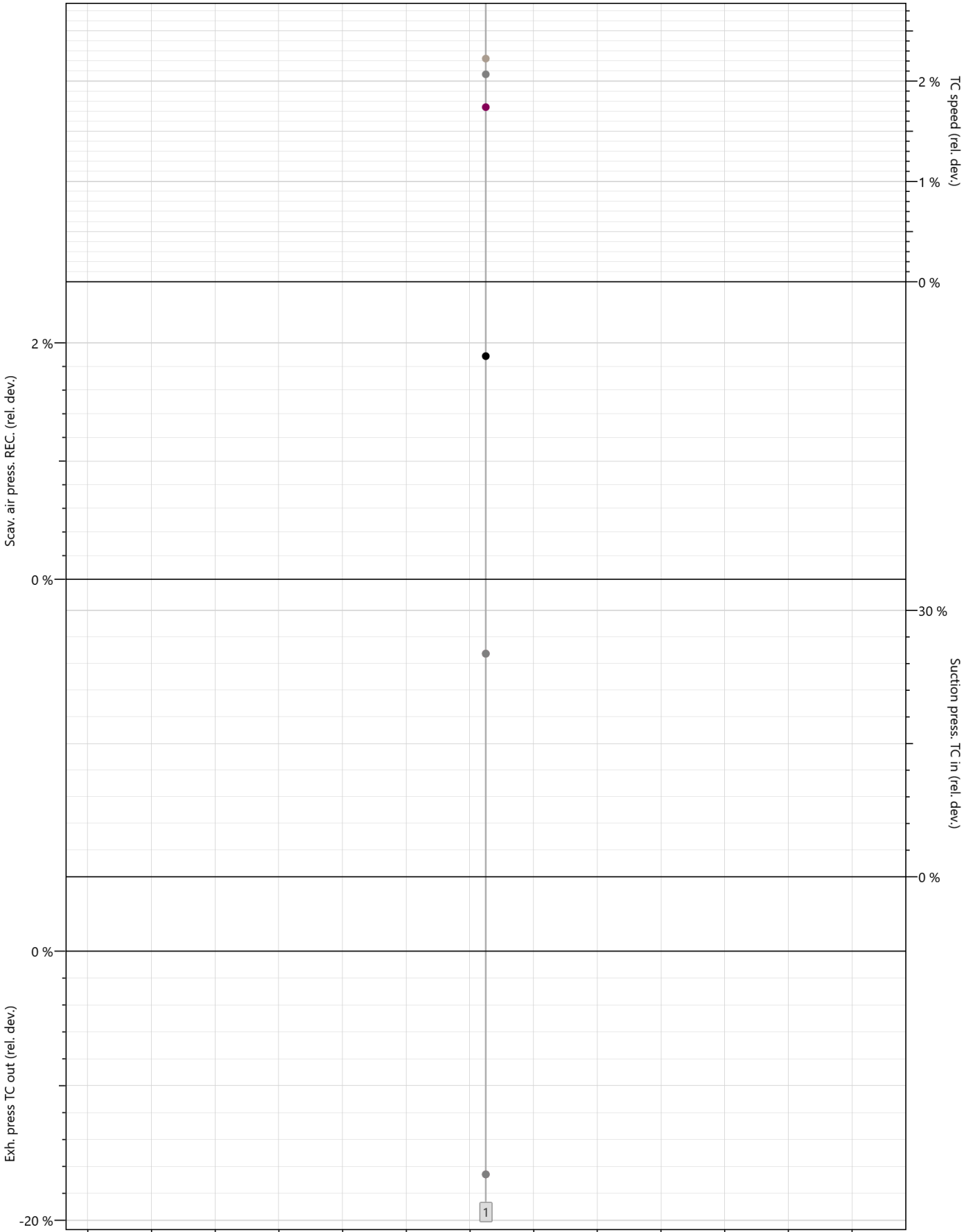
November 2025

● CYL 1	● CYL 2	● CYL 3	● CYL 4	● CYL 5	● CYL 6	● CYL 7	● CYL 8
● CYL 9	● CYL 10	- - AVG (CALC.)					

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Trend: TURBOCHARGER

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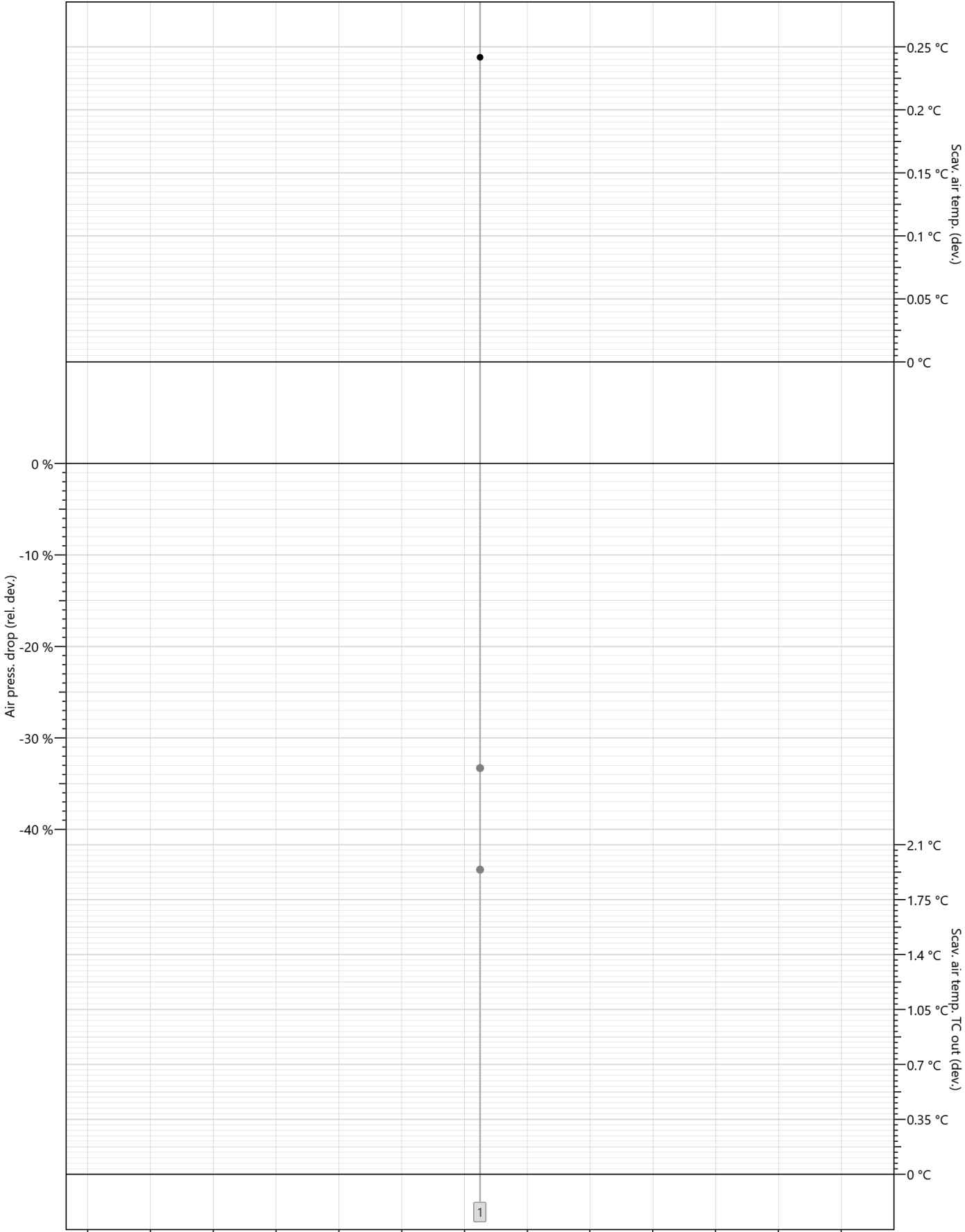
November 2025

● RECEIVER	● TC 1	● TC 2	● TC 3	— - AVG
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Trend: SCAV. AIR COOLER

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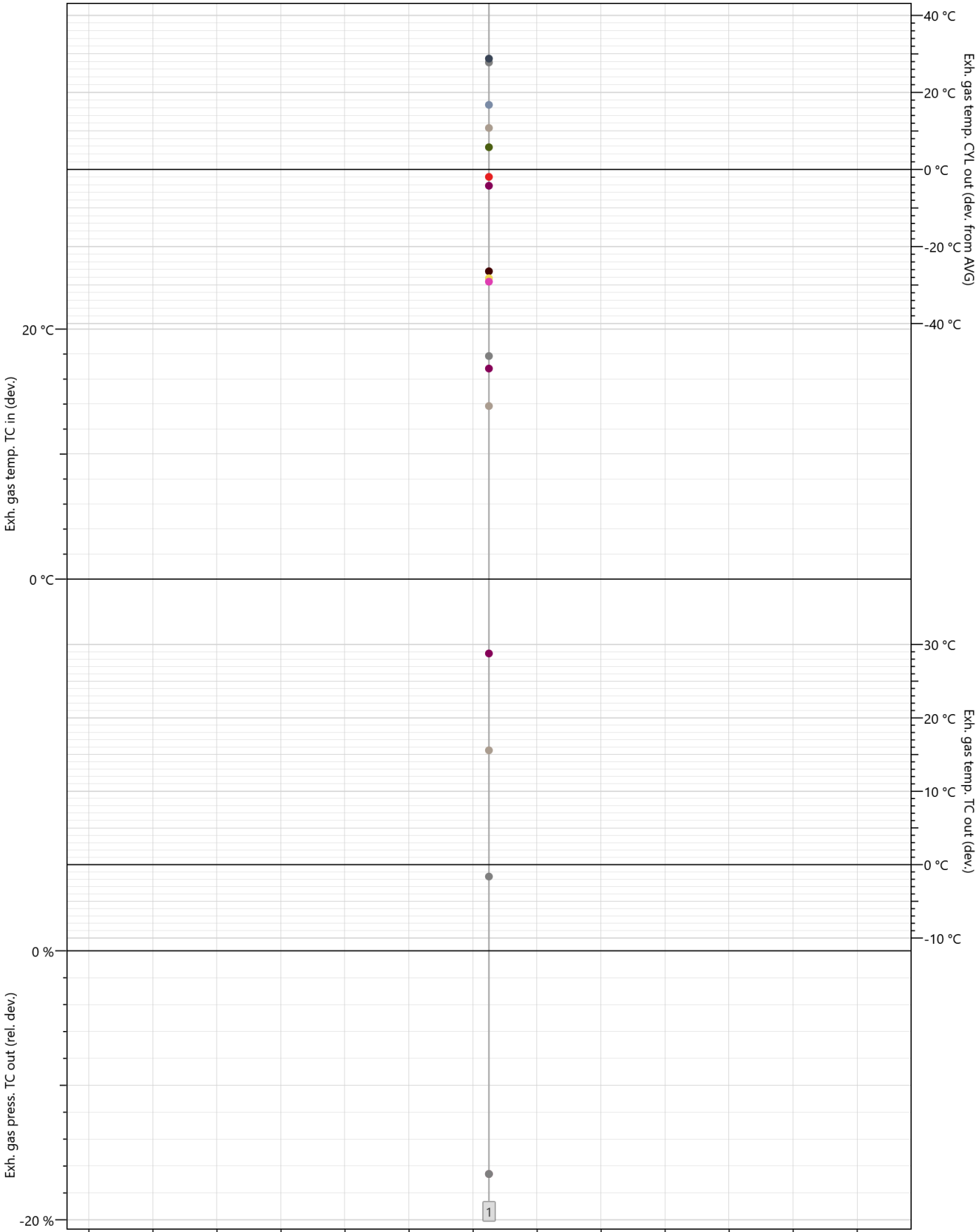
November 2025

TC 1	TC 2	TC 3	RECEIVER	AVG
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Trend: EXHAUST SYSTEM

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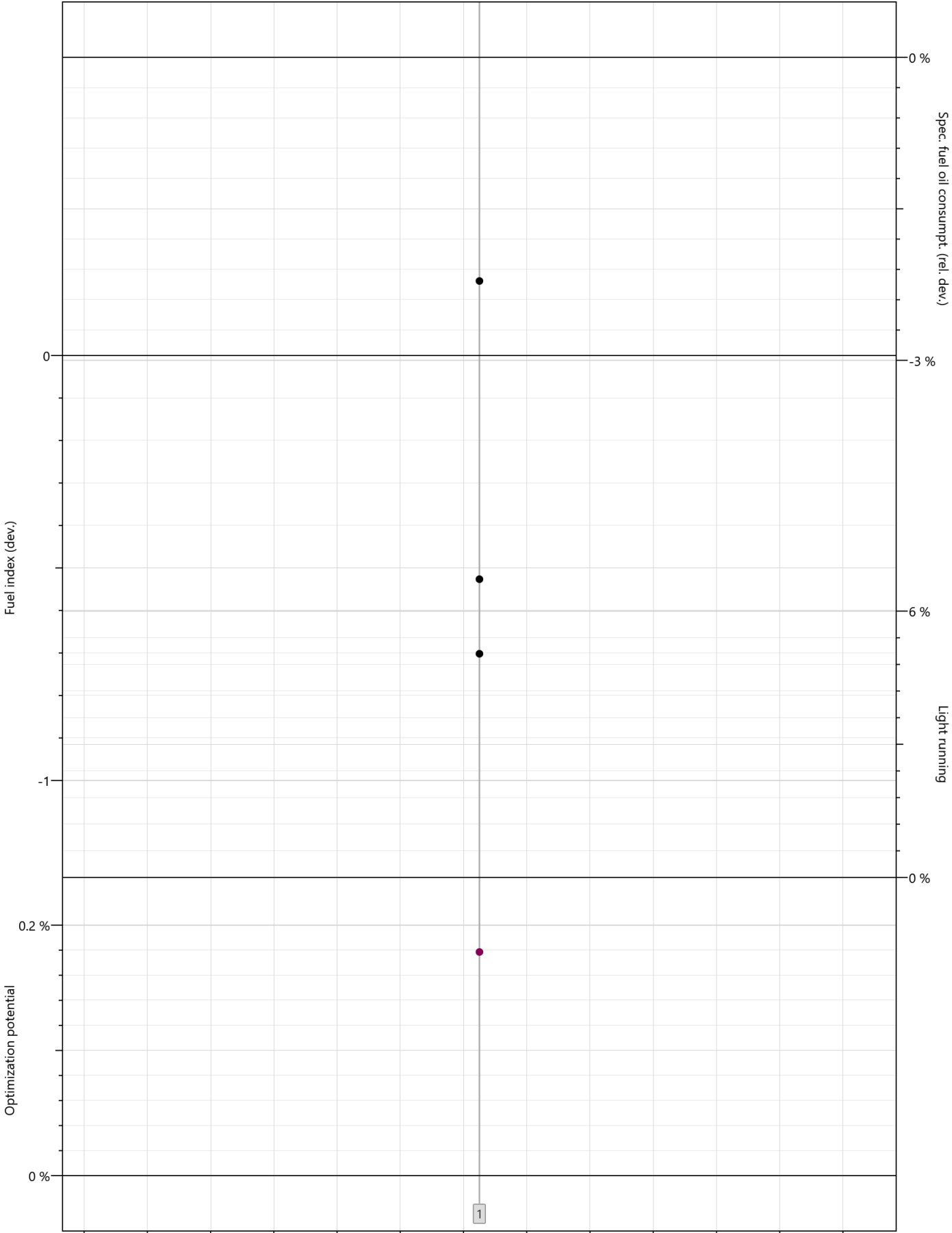
November 2025

CYL 1	CYL 2	CYL 3	CYL 4	CYL 5	CYL 6	CYL 7	CYL 8
CYL 9	CYL 10	TC 1	TC 2	TC 3	- AVG		

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Trend: CONSUMPTION

Installation	Equipment	Type	No.
APL HOUSTON	MAIN ENGINE	10S90ME-C	DML0105356



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Reading

Installation			Equipment			Type			No.																					
APL HOUSTON			MAIN ENGINE			10S90ME-C			DML0105356																					
General	Date and time of recording		11/11/2025 15 : 09 : 00		Sea water temp.		23.2 °C		SHIP SOG STW		17.00 17.60 kn		Voy. / title 04FLZW1MA																	
	ENGINE state		stable		Ambient air temp.		27.0 °C		Draft fwd aft		14.80 14.90 m		Record by 2E																	
	ENGINE running hours		69916 h		Engine room air temp.		35.0 °C		Draft mid Trim		14.85 0.10 m		Ackn. by CE																	
	Ambient humidity		rel 77.0 % abs 17.3 g/kg		Barom. pr. Eng. room		1.015 1.022 barA		Remarks		ENROUTE FROM SGSIN TO CGPNR																			
Power / Speed			CALCULATED		MEAS		ESTIMATED 24.3(E264X126)																							
			XPERT		TORSIOM.		MOP		by FI*rpm		by pscav		by TC rpm		by MEP		by rpm													
	ENGINE power estimated		20418 40.0		20660		20428 40.0		20185		20838		20347		20066															
			MCR		ENGINE		TORSIOMETER		MARGINS		MECH. EFFICIENCY																			
	ENGINE speed		84.0 77.4 % 65.0		Power kW		20660		Light running		5.0 %		Theoretic.		93.3 %															
Electronic ctrl	Running mode		Economy		ISO CORRECTED		MEASURED																							
	Governor mode		RPM		REF.		CALC.		ALL																					
	Fuel index ECU		57.5 56.9 54.0																											
	Applied fuel quality offset		0.0																											
Cylinder pressure	SOURCE: MAN CoCoS		ISO CORRECTED		MEASURED										SETPOINT															
	QUALITY Imported		REF.		CALC.		AVG.		CYL 1		CYL 2		CYL 3		CYL 4		CYL 5		CYL 6		CYL 7		CYL 8		CYL 9		CYL 10		MOP	
	Firing press. pmax		128.8 129.6		129.6		129.7		129.7		129.6		129.6		129.7		129.5		129.6											
	pmax deviation				ALL		0.1		0.0		0.1		0.0		0.0		0.0		0.1		-0.1		0.0							
	pmax offset				MOP SETTING:		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0							
	Compression press.		94.4 92.0		89.8		89.9		89.8		89.8		89.8		89.7		89.8		89.8		89.7		89.8							
	pcomp deviation				ALL		0.1		0.0		0.0		-0.1		0.0		0.0		0.0		-0.1		0.0							
	pcomp / pscav offset				MOP SETTING:		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0							
	Mean indicated press.		9.72		9.57		10.20		9.50		9.50		9.50		9.50		9.50		9.50		9.50									
	MIP deviation				0.63		-0.07		-0.07		-0.07		-0.07		-0.07		-0.07		-0.07		-0.07		-0.07							
	Fuel offset high				MOP SETTING:		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0							
	Fuel offset low				MOP SETTING:		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0							
	Power indicated		kW		SUM:		21501		2292		2134		2134		2134		2134		2134		2134		2134							
	Mean effective press. (MEP)		34.4 48.3		8.93		50.8 %		of MCR																					
	pmax-pcomp pcomp / pscav		- 39.8 46.2																											
Turbobcharger			ISO CORRECTED		MEASURED																									
			REF.		CALC.		AVG. / ENG.		TC 1		TC 2		TC 3																	
	PV (tot-tot)						1.93		1.93		1.93																			
	TC speed		7062 7204		7165		7146		7180		7169																			
	TC speed 298		7221 7029		7013		7010		7020		7009																			
	Turbocharger efficiency																													
	Surge margin																													
	Air mass flow		50.9 52.4		50.4		16.8		16.9		16.7																			
	Break spec.air cons.		8.9																											
	Exhaust gas mass flow		51.4 52.9		50.9		16.9		17.0		16.9																			
Scavenge air			ISO CORRECTED		MEASURED																									
	AUX. BLOWER operation		off		REF.		CALC.		AVG.		TC 1		TC 2		TC 3															
	Suction press.		16				20		20		20		20																	
	Press. drop accr. SAC		120		80		80		80		80																			
	Scav. air press. RECEIVER		0.97 1.01		0.95																									
	Scav.air press.setp. MOP																													
	Air temp. COMPR. in				35.3 34.0		36.0		36.0																					
	Press. COMPR. outlet *		0.97		0.92		0.92		0.92																					
	Scav. air temp. COMPR. out		113		115		115		115																					
	Scav. air temp. SAC out																													
	Scav. air temp. RECEIVER		35.8		34.0																									
	Cool. water temp. SAC in		29		32.0		32.0		32.0		32.0																			
	Cool. water temp. SAC out				36.7		36.0		37.0		37.0																			
	*measured at flange																													
Exhaust gas			ISO CORRECTED		MEASURED																									
			REF.		CALC.		AVG.		CYL 1		CYL 2		CYL 3		CYL 4		CYL 5		CYL 6		CYL 7		CYL 8		CYL 9		CYL 10			
	Exh. gas temp. CYL out		284 280		296		292		307		324		325		313		302		270		268		267		294					
	temp. deviation				ALL		-4		11		28		29		17		6		-26		-28		-29		-2					
	Exh. vlv. open offset				MOP SETTING:																									
	Exh. gas press. MANIFOLD		0.86 0.90		0.84		TC 1		TC 2		TC 3																			
	Exh. gas press. TC in *		0.84		0.84		0.84		0.84		0.84																			
	Exh. gas press. TC out		48		40		40		40		40																			
	Exh. gas temp. TC in		315 332		348		349		346		350																			
	Exh. gas temp. TC out		231 245		262		277		264		247																			
	Exh.wastegate (EGB) pos. area %		0 0 0 0																											
	*measured at flange																													

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Installation			Equipment				Type				No.							
APL HOUSTON			MAIN ENGINE				10S90ME-C				DML0105356							
Fuel oil			ISO CORRECTED		MEAS	Fuel oil properties:		ANALYS. ESTIMATE		Fuel oil system:		REF	MEAS	EXPECT.				
	Absolute consumption	m³/h ▾	REF.	CALC.	3.840	Viscosity	cSt	251	@ 50 °C	Viscosity ENGINE in	cSt	12.5	12.0	12.7				
	Specific FOC	g/kWh	168.9	165.2	172.2	Sulfur content	%	0.49	0.49	Temp. ENGINE in	°C	135	134					
	Light running correction	g/kWh	0.0			Water content	%	0.05	0.05	Temp. FLOWMETER	°C		89					
	Thrust bearing loss	g/kWh	-0.8			LCV	kJ/kg	41410	41410	Density FLOWM.	kg/m³		915.5					
	Specific FOC shoptest	g/kWh	168.0	164.3		Density	kg/m³	962.4	@ 15 °C	Price	USD/t							
						Source of analysis		Laboratory ▾										
Cyl. lub.			REFERENCE		MEAS	Cyl. lubr. system (Alpha):		REF	SET**	Cyl. lub. oil:		REF	ANALYS.					
	Absolute consumption	l/h ▾			17.0	Target feed rate@MCR	g/kWh	0.60	0.70	BN		40 - 70	70					
	Spec. lub oil consumpt. *	g/kWh			0.77	Target f.r.@current load	g/kWh	0.70	SET based	Density	kg/m³		940.0	@ 15 °C				
						** Acc.to OEM operation manuals and/or drainoil analysis					kg/m³		924	@ 40 °C				
Cylinder condition			REFERENCE		MEASURED													
			REF.			AVG.	ENGINE / CYL 1											
	Cool. water temp. CYL in	°C	75		80.0	80	CYL 2	CYL 3	CYL 4	CYL 5	CYL 6	CYL 7	CYL 8	CYL 9	CYL 10			
	Cool. water temp. CYL out	°C	90		85.1	85	84	86	85	86	84	87	84	84	86			
	Cooling water press. CYL in	bar			4.8	4.8	4.8	4.8	4.8	4.8	4.8	4.8	4.8	4.8	4.8			
	Coolant temp. PISTON in	°C			45	45	45	45	45	45	45	45	45	45	45			
	Coolant temp. PISTON out	°C			55	55	55	55	55	55	54	54	54	55	55			
	Scav. air temp. PIST. UNDERS.	°C			45.8	45	42	48	49	43	45	46	44	48	48			
Crankcase					MEASURED													
					AVG. / ENG.	BRG 1	BRG 2	BRG 3	BRG 4	BRG 5	BRG 6	BRG 7	BRG 8	BRG 9	BRG 10	BRG 11	BRG 12	BRG 13
	MAIN BEARING temp.	°C			53.0	52.3	55.5	54.7	54.3	54.6	52.4	50.4	51.7	52.3	54.5	55.3	50.4	50.3
	Temp. deviation	°C				-1	3	2	1	2	-1	-3	-1	-1	2	2	-3	-3
							CYL 1	CYL 2	CYL 3	CYL 4	CYL 5	CYL 6	CYL 7	CYL 8	CYL 9	CYL 10		
	CRANK PIN BEAR. temp	°C			51.4	51.0	51.0	51.0	51.0	52.0	54.0	52.0	51.0	50.0	51.0			
	Temp. deviation	°C				0	0	0	0	1	3	1	0	-1	0			
	X-HEAD BEAR. temp	°C			49.5	47.7	49.7	49.2	50.7	50.4	50.4	48.9	49.6	49.4	49.3			
Temp. deviation	°C				-2	0	0	1	1	1	-1	0	0	0				
Liner wall					MEASURED													
							CYL 1	CYL 2	CYL 3	CYL 4	CYL 5	CYL 6	CYL 7	CYL 8	CYL 9	CYL 10		
	LINER WALL temp. aft	°C																
	LINER WALL temp. fp	°C					125	129	132	141	125	129	116	122	122	132		
	LINER WALL temp. fwd	°C																
	LINER WALL temp. exh	°C			AVG.	124	124	129	129	130	127	122	126	118	126			
LINER WALL temp. AVG	°C			126														
TC/SAC media					MEASURED													
					AVG.	TC 1	TC 2	TC 3										
	Water press. SAC in	bar			2.8	2.8	2.8	2.8										
	Lub oil temp. TC in	°C			45	45	45	45										
Lub oil temp. TC out	°C			57	57	58	57											
Engine media			MEAS				MEASURED											
	Lub oil temp ENGINE in	°C			45.0	Fuel oil press. ENGINE in		bar	7.50									
	Lub oil temp. THRUST SEGM.	°C			56.0	Fuel oil press. ENGINE out		bar	7.20									
	Lub oil press. SYSTEM OIL	bar			2.42	Fuel oil press. FILTER in		bar	8.0									
	Lub oil press. COOLING OIL	bar			2.35	Fuel oil press. FILTER out		bar	7.5									
	Lub oil press. TURB. OIL	bar			2.20	Hydraulic oil press. FILTER in		bar	2.5									
	Lub oil temp. TURB. OIL in	°C			44.63	Hydraulic oil press. FILTER out		bar	2.4									
	Lub oil temp. TURB. OIL out	°C			57.47	Hydraulic oil press. MAIN		bar	229.0									
	Axial vibration	mm			0.71	Sealing air press.		bar		PUMP 2	PUMP 3	PUMP 4	PUMP 5					
						Hydr. oil PUMP swash plate pos.		%	50.0	58.0	58.0							
	Plant					MEAS												
Exh. gas differential press. ECONOMIZER		mmWG			15.00													